

Dynamic ISF: can sensitivity be derived from closed-loop data?

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This is the corrected evidence summary for the historical feasibility work. The original physiological constants and dosing recommendations are withdrawn. See DYNAMIC-ISF-AUDIT-2026-09.md for the full review.

Question

Can a person's physiological insulin sensitivity factor be calculated from routine closed-loop glucose, IOB and delivery records well enough to replace an entered ISF?

Short answer

Not with the estimators tested here.

The analyses can calculate useful outcome proxies. They can compare equations, describe entered settings and test prediction error inside a loop model. They do not isolate the glucose effect of absorbed insulin from new insulin delivery, controller response, basal need, endogenous glucose and residual carbohydrate action. The resulting coefficient should not be called measured ISF.

What the historical programme measured

The work used 138 people from open-source AID cohorts and a series of related analyses. The table below describes the original outputs using their corrected interpretation.

analysis	historical result	audited interpretation
short-window stability	14-day estimates varied markedly between blocks	the proxy is too unstable for direct dosing calibration
variance decomposition	most short-term variation appeared residual	this may be estimator noise and controller feedback, not biological sensitivity variation
shrinkage	longer windows and population shrinkage improved repeatability	repeatability of a proxy does not establish physiological validity
outcome association	the proxy was associated with existing	this is evidence of endogeneity and makes

analysis	historical result	audited interpretation
overnight restriction	low-glucose outcomes the same bias remained overnight	causal dosing use unsafe removing obvious meals did not identify insulin effect
outcome-based adjustment	bounded outcome tuning reduced the recorded hypo signature	this is a separate control strategy, not validation of the ISF proxy

The original work described a square-root TDD relationship and constants near 355 for entered profiles and 145 for the change-in-IOB proxy. Those numbers remain reproducible descriptions of their targets. They are not physiological constants and should not be used as dosing defaults.

Why the estimator does not identify ISF

The historical regression related glucose change to change in modelled IOB during a fasting window. That seems intuitive, but three quantities are mixed together:

1. insulin already active at the beginning of the window;
2. new bolus and temporary-basal insulin delivered during the window;
3. insulin leaving the modelled IOB curve as time passes.

The controller changes the second quantity in response to glucose and trend. The exposure is therefore selected by the same process that is trying to correct the outcome. A coefficient from that regression cannot, by itself, be interpreted as glucose lowering per unit of absorbed insulin.

The later action-balance estimator does not remove the problem. Its IOB is generated by an assumed insulin-action model, and its delivery term includes suggested SMB and temporary basal rather than confirmed enacted delivery in every record. It is a second observational proxy, not an independent ground truth.

What remains supported

The equation implementations are auditable

The Python implementation reproduces the intended v1 and author-confirmed v2 equations. V2 uses $\ln(\text{BG}/\text{divisor})$ without +1 and floors glucose at $\text{divisor}+1$. For the standard rapid-acting configuration, including NovoRapid, that is $\text{divisor}/\text{floor } 75/76$.

Static ISF is competitive in the tested prediction model

In selected overnight windows, a person's tuned static ISF predicted the realised end glucose about as well as the loop's own calibrated ISF and better than either dynamic equation. This is a within-loop prediction result. It does not measure the biological ISF level.

The tested data do not support a glucose multiplier

With contiguous time-fold validation, the best fitted glucose exponent remained zero and the glucose term improved median absolute error by 0.0 mg/dL. Within these data and this prediction model, adding glucose-dependent sensitivity did not help.

Residual carbohydrate is a credible explanation

Separate post-COB analyses found lower apparent insulin effect soon after recorded COB cleared. This is consistent with carbohydrate absorption continuing after the controller's COB model has ended. It supports the hypothesis that some apparent Dynamic ISF behaviour compensates for an imperfect carbohydrate model. It does not prove that every successful use of Dynamic ISF has that cause.

What is withdrawn

The audit withdraws these earlier claims:

- that physiological ISF follows $1/\sqrt{TDD}$;
- that constants derived from the change-in-IOB proxy are measured ISF constants;
- that a per-person physiological ISF can be safely calibrated from these closed-loop records;
- that the Diabeloop-shaped glucose multiplier is supported by this cohort;
- that the historical Tier 1 or Tier 2 v-next forms are ready for live dosing.

The entered profile ISF remains a description of how people configured their systems. It should not be presented as a biological measurement either.

Practical conclusion

Routine closed-loop records are valuable for equation replay, software verification, prediction testing and hypothesis generation. They are not sufficient on their own to recover a causal ISF without a design that separates insulin exposure from controller response and carbohydrate effects.

The next step is prospective validation or a carefully identified natural experiment using:

- confirmed bolus and temporary-basal delivery;
- a specified insulin-action model and sensitivity analyses across plausible models;
- explicit carbohydrate and rescue-treatment handling;
- non-overlapping evaluation windows;
- a pre-specified static comparator;
- safety outcomes as well as prediction error.

Until then, the historical fitted constants and square-root equation are research findings, not dosing recommendations.

Reproducibility

- Historical phases: `inv008/phase{1,2,3,5,6}_.py`
- Same-window comparison: `inv008/head_to_head.py`
- Glucose-shape comparison: `inv008/gradient_isf_fit.py`
- Post-COB analysis: `inv008/post_cob_isf.py`
- Audit: `DYNAMIC-ISF-AUDIT-2026-09.md`